



Short Communication

T-linear specific heat in pressurized and magnetized Shastry-Sutherland Mott insulator $\text{SrCu}_2(\text{BO}_3)_2$

Jing Guo, Pengyu Wang, Cheng Huang, Chengkang Zhou, Wenshan Hong, Menghan Song, Xintian Chen, Ting-Tung Wang, Shu Cai, Jinyu Zhao, Jinyu Han, Yazhou Zhou, Qi Wu, Shiliang Li, Zi Yang Meng, Liling Sun

PII: S2095-9273(26)01157-6
DOI: <https://doi.org/10.1016/j.scib.2026.09.060>
Reference: SCIB 4408

To appear in: *Science Bulletin*

Received Date: 29 April 2026
Revised Date: 29 April 2026
Accepted Date: 16 September 2026

Please cite this article as: J. Guo, P. Wang, C. Huang, C. Zhou, W. Hong, M. Song, X. Chen, T-T. Wang, S. Cai, J. Zhao, J. Han, Y. Zhou, Q. Wu, S. Li, Z.Y. Meng, L. Sun, *T*-linear specific heat in pressurized and magnetized Shastry-Sutherland Mott insulator $\text{SrCu}_2(\text{BO}_3)_2$, *Science Bulletin* (2026), doi: <https://doi.org/10.1016/j.scib.2026.09.060>

This is a PDF of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability. This version will undergo additional copyediting, typesetting and review before it is published in its final form. As such, this version is no longer the Accepted Manuscript, but it is not yet the definitive Version of Record; we are providing this early version to give early visibility of the article. Please note that Elsevier's sharing policy for the Published Journal Article applies to this version, see: <https://www.elsevier.com/about/policies-and-standards/sharing#4-published-journal-article>. Please also note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

***T*-linear specific heat in pressurized and magnetized Shastry-Sutherland Mott insulator $\text{SrCu}_2(\text{BO}_3)_2$**

Jing Guo^{a,b,1}, Pengyu Wang^{a,b,1}, Cheng Huang^{c,d,1}, Chengkang Zhou^{c,d,1}, Wenshan Hong^{a,b,1}, Menghan Song^{c,d}, Xintian Chen^{a,b}, Ting-Tung Wang^{c,d}, Shu Cai^{a,e}, Jinyu Zhao^{a,b}, Jinyu Han^{a,b}, Yazhou Zhou^a, Qi Wu^a, Shiliang Li^{a,b}, Zi Yang Meng^{c,d,*}, Liling Sun^{a,b,e,*}

^a Beijing National Laboratory for Condensed Matter Physics and Institute of Physics, Chinese Academy of Sciences, Beijing 100190, China

^b University of Chinese Academy of Sciences, Beijing 100049, China

^c Department of Physics and HK Institute of Quantum Science & Technology, The University of Hong Kong, Hong Kong, China

^d State Key Laboratory of Optical Quantum Materials, The University of Hong Kong, Hong Kong, China

^e Center for High Pressure Science & Technology Advanced Research, Beijing 100193, China

* Corresponding authors.

E-mail addresses: zymeng@hku.hk (Zi Yang Meng), llsun@iphy.ac.cn (Liling Sun).

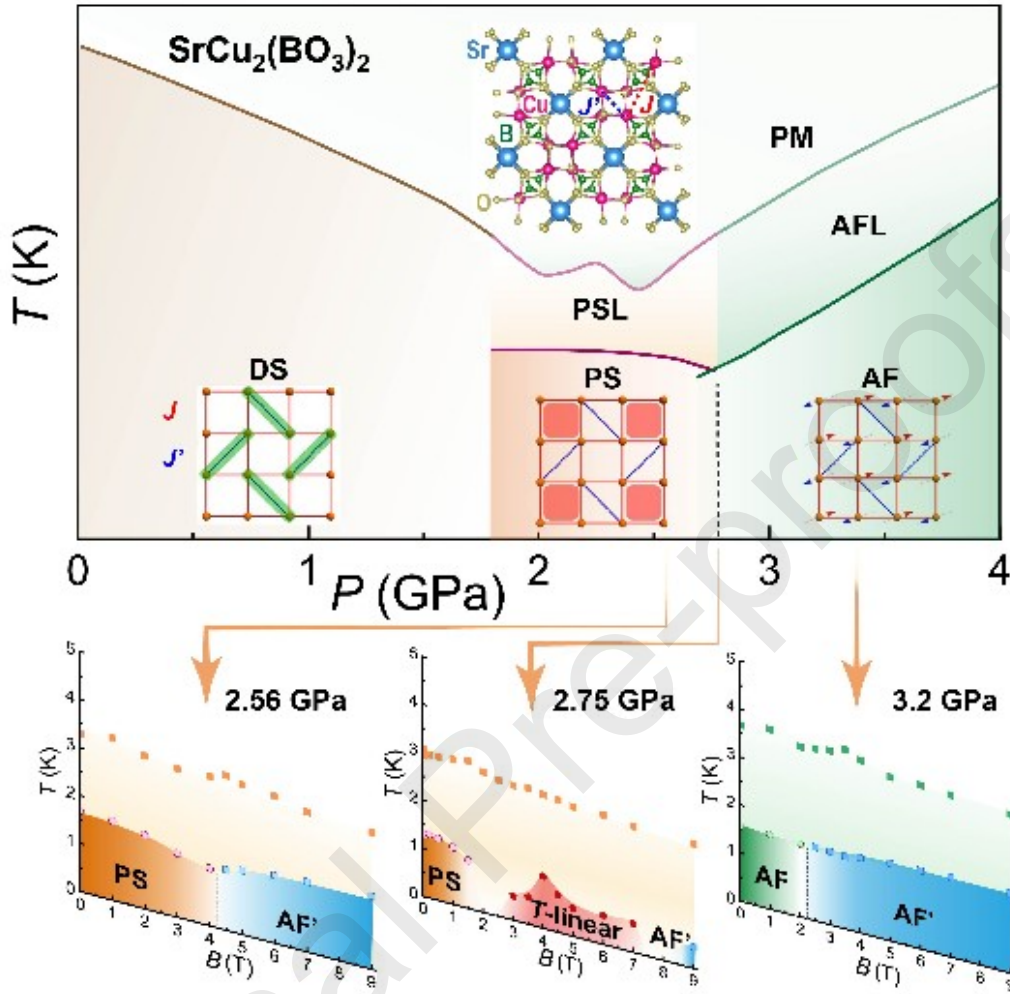
¹ These authors contributed equally to this work.

Received: 29-Apr-2026

Revised: 29-Apr-2026

Accepted: 16-Sep-2026

Graphical abstract



The search for deconfined quantum criticality and fractionalized excitations beyond the Landau-Ginzburg-Wilson paradigm has been a central theme in frustrated quantum magnetism [1–4]. The Shastry-Sutherland Mott insulator $\text{SrCu}_2(\text{BO}_3)_2$ has long been regarded as a promising platform due to its proximity to competing plaquette-singlet (PS) and antiferromagnetic (AF) phases [5–11]. These two phases break distinct symmetries, and their transition has been proposed as a candidate for hosting exotic criticality or even a quantum spin liquid [9,12–14]. However, recent high-pressure measurements revealed that the zero-field PS-AF transition is first-order, seemingly precluding deconfined critical behavior [10].

Previous high-field studies explored pressures up to 2.4 GPa [11]. Here we push the pressure limit up to 3.2 GPa to access unconventional gapless excitations in this system. By performing specific-heat measurements under combined high pressure, low temperature, and magnetic field, we uncover a robust and universal T -linear specific heat over a wide magnetic-field range near the pressure corresponding to the zero-field PS-AF transition. Such metallic-like thermodynamic behavior is highly unusual for a Mott insulator, where charge degrees of freedom remain gapped. Optical measurements confirm that the system remains insulating across the investigated pressure and field

ranges, suggesting that the observed linear-in- T contribution does not originate from itinerant charge carriers. In addition, the wide parameter ranges over which the behavior is observed, and its reproducibility make extrinsic origins, such as disorder or glassy effects, unlikely as a dominant mechanism.

To place these experimental observations in a broader theoretical context, we perform quantum many-body simulations of the Shastry-Sutherland model under magnetic fields using the exponential thermal tensor renormalization group (XTRG) method [15]. The calculations show an enhanced low-energy thermodynamic response near the PS-AF transition, consistent qualitatively with the observed low-temperature behavior. These results provide theoretical context for the observed gapless thermodynamics, but do not by themselves determine its microscopic origin. Fractionalized excitations remain one possible interpretation [16,17], alongside other low-energy modes associated with competing phases. Given the current experimental and theoretical evidence, a definitive identification of the underlying microscopic mechanism remains open.

Taken together, these results reveal a previously unexplored regime in $\text{SrCu}_2(\text{BO}_3)_2$, where pressure and magnetic field cooperatively generate robust gapless thermodynamic behavior near a first-order phase boundary. This finding highlights the richness of the phase diagram and calls for further investigations into the nature of low-energy excitations in frustrated quantum magnets.

To obtain these thermodynamic signatures, low-temperature heat-capacity measurements under combined high pressure and magnetic field were performed using an AC calorimetry technique. A piston-cylinder pressure cell was employed to achieve pressures up to ~ 3.2 GPa (Fig. S1 online), temperatures down to ~ 0.4 K, and magnetic fields up to 14 T. A glycerin/water mixture (3:2) was used as the pressure-transmitting medium to ensure quasi-hydrostatic conditions, and the applied pressure was determined from the superconducting transition of Pb placed alongside the sample.

For AC calorimetry, a resistive heater was attached to the sample to generate temperature oscillations driven by an AC current, while a thermocouple mounted on the opposite side detected the resulting temperature modulation. The heat capacity was obtained from the ratio between the input power and the temperature oscillation amplitude under quasi-adiabatic conditions, with the excitation frequency optimized during cooling.

To account for background contributions from the pressure medium and surrounding components, separate measurements and calibrations were performed, allowing an estimation of the addenda contribution to the total heat capacity. Details of the experimental setup, calibration procedures, and data analysis are provided in the Supplementary materials (Sections S2 and S3, Figs. S2–S4 online)

With the measurement and calibration procedures established, we then performed high-pressure heat-capacity measurements on the sample under combined pressure and magnetic field conditions. The obtained heat-capacity data are presented in Fig. 1 and Figs. S5–S7 (online). At 2.3 GPa and fields below 5.5 T (Fig. 1a and Fig. S5a online), the critical temperature of the PS phase (T_{PS}) shifts to lower temperatures as the magnetic field increases, consistent with previous results [8,9]. The intensity of the PS peak diminishes with an increasing field and becomes undetectable when the magnetic field reaches 5.5 T. Afterwards, a new peak associated with a magnetic-field-induced antiferromagnetic phase (AF') emerges [8,9], with its transition temperature $T_{\text{AF}'}$

increasing with field (Fig. 1b), consistent with nuclear magnetic resonance (NMR) studies at comparable pressure [9], indicating that the thermodynamic features we observe originate from the same underlying microscopic spin reorganization. At 2.56 GPa, the PS-AF' transition sharpens into a pronounced first-order transition near 1.2 K and 4.2 T (as shown in Fig. 1c and d, Fig. S5b online). The PS anomaly persists just below 4 T but is abruptly replaced by a strong AF' peak at 4.5 T, signaling the collapse of the PS phase and the onset of AF'. Compared with 2.3 GPa, the $T_{AF'}$ peaks are sharper, indicating a more distinct transition under increased pressure.

At higher pressures, distinct behavior develops. At 2.68 GPa, the PS state is suppressed by a very low magnetic field (~ 1 T, see Fig. 1e). Between 1 and 4 T, no peak corresponding to the PS or AF' state is detected down to 0.4 K. In the magnetic field range of 1.5–4.0 T, the specific heat C exhibits a T -linear behavior below 1.5 K with an intercept value of around $1.1\text{--}1.4\text{ J K}^{-2}\text{ mol}^{-1}$ for C/T (Fig. 1e and f, labeled by the dashed line and the yellow shadow), making a notable difference from the phase diagrams at 2.3 and 2.56 GPa between the PS and AF' phases. This T -linear behavior disappears above 4 T, where the AF' peaks start to appear.

Upon increasing the pressure further to 2.75 GPa, corresponding to the pressure of the zero-field first-order PS-AF transition [10], the T -linear behavior exists in an even broader magnetic range of 3–7 T, as labeled by dashed lines and yellow shadows in Fig. 1g and h, followed by the appearance of the AF' phase starting at a higher field (9 T), indicating that the system changes significantly under pressure and magnetic field. The linear specific heats C/T exhibit an intercept universally around $1.1\text{--}1.6\text{ J K}^{-2}\text{ mol}^{-1}$, indicating a remarkable universality of the low-energy thermodynamic response. The reproducibility of the T -linear regime is further supported by magnetic-field-dependent heat-capacity measurements on an independent sample at 2.92 GPa, which reveal the same field-induced evolution from AF to the T -linear regime and subsequently to AF' (Fig. S7 online).

The insulating nature of the system is preserved throughout this regime, as confirmed by optical measurements (Fig. S8 online), indicating that charge degrees of freedom remain frozen, and the system remains a Mott insulator. Moreover, the robustness and universality of the linear term across different pressures and fields make extrinsic origins such as disorder or glassy behavior unlikely. As in those cases, the C/T could give rise to very different low-temperature behavior (for example, different intercepts) at different magnetic fields and pressures. Instead, these observations point to the emergence of unconventional low-energy spin excitations. For convenience, we refer to this regime as a gapless intermediate state.

At the pressure of 2.9 GPa, according to the previous zero-field results [10], the zero-field ground state has changed to the pressure-induced AF phase. As shown in Fig. 1i, the phase transition temperature of the AF phase decreases with increasing magnetic fields below 2 T. At this pressure, the T -linear range of the specific heat shrinks significantly, as we can observe within a short range of 2.5–4.0 T, but still with an intercept universally around $1.2\text{ J K}^{-2}\text{ mol}^{-1}$, as also indicated by the dashed line in Fig. 1i and j. When the magnetic field is larger than 4.25 T, a clear low-temperature AF' peak appears (Fig. 1j).

At 3.2 GPa, no T -linear behavior is observed at any field, and the transition changes to the AF-AF' type (Fig. 1k and l). The transition temperature of the peaks related to AF decreases slightly with increasing magnetic field below 2.0 T, while the peak intensity weakens and broadens. Above 2.0 T, the AF' phase emerges and its peak shifts to higher temperatures with increasing magnetic field, accompanied by enhanced peak intensity (Fig. 1l). This evolution suggests that the gapless intermediate state is

most prominent near the first-order PS-AF boundary and is progressively suppressed away from it.

To visualize the evolution of the phase diagram, we construct color maps of C/T as functions of temperature and magnetic field (Fig. S6 online). At low pressure (2.3 and 2.56 GPa), the system evolves directly from the PS to AF' phase with increasing field (Fig. S6a and b online). In contrast, near the PS-AF boundary (2.68–2.9 GPa), a finite-field window with temperature-linear specific heat clearly emerges, reaching its maximum extent near 2.75 GPa and gradually shrinking at higher pressures (Fig. S6c–e online). At 3.2 GPa, the temperature-linear regime disappears entirely, and the system exhibits only a transition or a crossover between AF and AF' phases, indicating the suppression of the intermediate gapless state at high pressure (Fig. S6f online).

The combined pressure-field phase diagram is summarized in Fig. 2a–g. The temperature-linear specific heat regime emerges under a magnetic field in the vicinity of the first-order PS-AF transition, indicating a strong sensitivity of the low-energy thermodynamic behavior to external tuning parameters. The low-temperature phase diagram as a function of magnetic field and pressure is further illustrated in Fig. 2h, highlighting the extended region where the temperature-linear specific heat is observed. The emergence of such a robust gapless thermodynamic regime in such an insulating system suggests unconventional low-energy magnetic excitations beyond conventional symmetry-breaking orders and may be indicative of fractionalized spin degrees of freedom.

We further perform quantum many-body simulations using the XTRG method [15] to show that the Shastry-Sutherland lattice can also produce T -linear specific heat under a magnetic field near the PS-AF boundary (See Supplementary materials, Section S6, Fig. S9 online). In addition, zero-field calculations in both the dimer-singlet (DS) and PS phases show rapidly vanishing specific heat at low temperatures (Fig. S10 online), consistent with gapped excitations, in good agreement with the experimental results (Fig. 1). The comparison between finite-field and zero-field results highlights the role of magnetic field in inducing gapless thermodynamic behavior across the PS-AF boundary (Fig. 2) and suggests the presence of unconventional magnetic excitations induced by the combined effects of pressure and magnetic field [16,17]. However, the microscopic origin of such excitations remains to be established.

Taken together, the experimental and theoretical results reveal a robust and nearly universal temperature-linear specific heat over a wide range of pressure and magnetic fields, with the maximum field window occurring near the zero-field first-order PS-AF transition (Fig. 2h). The observed gapless response may relate to unconventional low-energy spin excitations. Similar signatures of such excitations have been reported in the inelastic neutron scattering in the Kagome antiferromagnet $\text{YCu}_3(\text{OD})_6\text{Br}_2[\text{Br}_{(1-x)}(\text{OD})_x]$ [18,19], triangular antiferromagnets $\text{YbZn}_2\text{GaO}_5$ [16,17], and KYbSe_2 [20], and the putative quantum oscillation at the $1/9$ magnetic plateau of the Kagome material [21]. In this context, $\text{SrCu}_2(\text{BO}_3)_2$ provides a distinct platform where pressure and magnetic field can be used as effective tuning parameters to access this regime.

The overall consistency between experiment and tensor-network simulation further supports a scenario in which the gapless behavior arises from the interplay of frustration and external tuning parameters. However, as a bulk thermodynamic probe, specific heat alone cannot resolve the momentum- or spin-dependent characteristics of these excitations. While the present results are compatible with scenarios involving fractionalized spin excitations, a definitive identification of the microscopic mechanism remains an open question. Complementary experimental probes, such as neutron

scattering, magnetic torque, susceptibility, and NMR, will be essential to clarify the nature of these excitations and their possible connections to deconfined quantum states.

Conflict of interest

The authors declare that they have no conflict of interest.

Acknowledgments

This work was supported by the National Key Research and Development Program of China (2021YFA1401800, 2022YFA1403900, 2022YFA1403400, 2025YFA1411500, and 2021YFA1400400), the National Natural Science Foundation of China (U2032214, 12474054, 12122414, 12104487, and 12004419), the Strategic Priority Research Program (B) of the Chinese Academy of Sciences (XDB25000000 and XDB33000000), the Chinese Academy of Sciences Superconducting Research Project (SCZX-0101), the Synergetic Extreme Condition User Facility (SECUF, <https://cstr.cn/31123.02.SECUF>), the K. C. Wong Education Foundation (GJTD-2020-01), and the Open Research Fund of the Pulsed High Magnetic Field Facility (WHMFC2024006), Huazhong University of Science and Technology. Jing Guo and Shu Cai are grateful for support from the Youth Innovation Promotion Association of the CAS (2019008) and the China Postdoctoral Science Foundation (E0BK111). Cheng Huang, Ting-Tung Wang, Chengkang Zhou and Ziyang Meng acknowledge the support from the Research Grants Council of Hong Kong (AoE/P-701/20, 17309822, C7037-22GF, 17302223, and 17301924), the ANR/RGC Joint Research Scheme sponsored by the RGC of Hong Kong and French National Research Agency (A_HKU703/22), and the HKU Seed Funding for Strategic Interdisciplinary Research “Many-body paradigm in quantum moiré material research”. We thank the HPC2021 system under the Information Technology Services, University of Hong Kong, as well as the Beijing PARATERA Tech CO., Ltd. (<https://cloud.paratera.com>) for providing HPC resources that have contributed to the research results reported in this paper.

Author contributions

Ziyang Meng, Liling Sun, and Jing Guo initiated the work. Liling Sun and Ziyang Meng supervised experimental and theoretical aspects, respectively. Wenshan Hong and Shiliang Li grew the $\text{SrCu}_2(\text{BO}_3)_2$ single crystals. Jing Guo and Pengyu Wang conducted high-pressure heat capacity measurements with the help of Xintian Chen, Shu Cai, Jinyu Zhao, Jinyu Han, and Yazhou Zhou. Ziyang Meng, Cheng Huang, Chengkang Zhou, Menghan Song, and Ting-Tung Wang performed theoretical calculations. Liling Sun, Ziyang Meng, Jing Guo, Cheng Huang, Ting-Tung Wang, Chengkang Zhou, Shiliang Li, and Qi Wu carried out data analysis. Ziyang Meng, Liling Sun, Jing Guo, Cheng Huang, Chengkang Zhou, Shiliang Li, Qi Wu, and Ting-Tung Wang wrote the manuscript with the efforts of all authors.

References

- [1] Landau LD, Lifshitz EM. Chapter IX—Solutions. In: Statistical Physics. 3rd ed. Oxford: Butterworth-Heinemann; 1980. p. 263–304.
- [2] Wilson KG. The renormalization group and critical phenomena. *Rev Mod Phys* 1983; 55, 583–600.

- [3] Senthil T, Vishwanath A, Balents L, et al. Deconfined quantum critical points. *Science* 2004; 303: 1490–1494.
- [4] Senthil T, Balents L, Sachdev S, et al. Quantum criticality beyond the Landau-Ginzburg-Wilson paradigm. *Phys Rev B* 2004; 70: 144407.
- [5] Shastry BS, Sutherland B. Exact ground state of a quantum mechanical antiferromagnet. *Physica B+C* 1981; 108: 1069–1070.
- [6] Zayed ME, Rüegg Ch, Larrea J J, et al. 4-spin plaquette singlet state in the Shastry-Sutherland compound $\text{SrCu}_2(\text{BO}_3)_2$. *Nat Phys* 2017; 13: 962–966.
- [7] Guo J, Sun G, Zhao B, et al. Quantum phases of $\text{SrCu}_2(\text{BO}_3)_2$ from high-pressure thermodynamics. *Phys Rev Lett* 2020; 124: 206602.
- [8] Jiménez JL, Crone SPG, Fogh E, et al. A quantum magnetic analogue to the critical point of water. *Nature* 2021; 592: 370–375.
- [9] Cui Y, Liu L, Lin H, et al. Proximate deconfined quantum critical point in $\text{SrCu}_2(\text{BO}_3)_2$. *Science* 2023; 380: 1179–1184.
- [10] Guo J, Wang P, Huang C, et al. Deconfined quantum critical point lost in pressurized $\text{SrCu}_2(\text{BO}_3)_2$. *Commun Phys* 2025; 8: 75.
- [11] Shi ZZ, Dissanayake S, Corboz P, et al. Discovery of quantum phases in the Shastry-Sutherland compound $\text{SrCu}_2(\text{BO}_3)_2$ under extreme conditions of field and pressure. *Nat Commun* 2022; 13: 2301.
- [12] Xi N, Chen HY, Xie ZY, et al. Plaquette valence bond solid to antiferromagnet transition and deconfined quantum critical point of the Shastry-Sutherland model. *Phys Rev B* 2023; 107: L220408.
- [13] Lee JY, You YZ, Sachdev S, et al. Signatures of a deconfined phase transition on the Shastry-Sutherland lattice: applications to quantum critical $\text{SrCu}_2(\text{BO}_3)_2$. *Phys Rev X* 2019; 9: 041037.
- [14] Yang JW, Sandvik AW, Wang L. Quantum criticality and spin liquid phase in the Shastry-Sutherland model. *Phys Rev B* 2022; 105: L060409.
- [15] Song M, Zhou C, Huang C, et al. Thermodynamics of Shastry-Sutherland model under magnetic field. *arXiv*: 2026;2602.11589.
- [16] Bag R, Xu S, Sherman NE, et al. Evidence of Dirac quantum spin liquid in $\text{YbZn}_2\text{GaO}_5$. *Phys Rev Lett* 2024; 133: 266703.
- [17] Wu HCH, Pratt FL, Huddart BM, et al. Spin dynamics in the Dirac $\text{U}(1)$ spin liquid $\text{YbZn}_2\text{GaO}_5$. *Phys Rev Lett* 2025; 135: 046704.
- [18] Zeng Z, Zhou C, Zhou H, et al. Spectral evidence for Dirac spinons in a kagome lattice antiferromagnet. *Nat Phys* 2024; 20: 1097–1102.
- [19] Han L, Zeng Z, Long M, et al. Spin excitations arising from anisotropic Dirac spinons in $\text{YCu}_3(\text{OD})_6\text{Br}_2[\text{Br}_{0.33}(\text{OD})_{0.67}]$. *Phys Rev B* 2025; 112: 045114.
- [20] Scheie AO, Ghioldi EA, Xing J, et al. Proximate spin liquid and fractionalization in the triangular antiferromagnet KYbSe_2 . *Nat Phys* 2024; 20: 74–81.
- [21] Zheng G, Zhu Y, Chen K, et al. Unconventional magnetic oscillations in a kagome Mott insulator. *Proc Natl Acad Sci USA* 2025; 122: e2421390122.

Figure captions

Fig. 1. Evolution of the specific heat capacity C/T of $\text{SrCu}_2(\text{BO}_3)_2$ as a function of magnetic field and temperature for pressures ranging from 2.3 to 3.2 GPa. C/T vs. T measured at 2.3 GPa (a, b), 2.56 GPa (c, d), 2.68 GPa (e, f), 2.75 GPa (g, h), 2.9 GPa (i, j), and 3.2 GPa (k, l). T_{PS} , $T_{\text{AF}'}$, and T_{AF} stand for the transition temperature of the plaquette singlet (PS) state, magnetic-field-induced anti-ferromagnetic (AF') state, and pressure-induced anti-ferromagnetic (AF) state, respectively. T -linear specific heat regions are denoted by dashed lines and yellow shadows.

Fig. 2. The pressure and magnetic field phase diagram of $\text{SrCu}_2(\text{BO}_3)_2$. (a) The zero-field pressure-temperature (P - T) phase diagram. Here, the acronyms DS, PS, AF, and PM stand for dimer singlet, plaquette singlet, the pressure-induced anti-ferromagnetism, and para-ferromagnetism, respectively. PSL and AFL stand for the correlation forming of the corresponding orders of PS and AF, respectively, with L stands for liquid. Adapted from Ref. [10] under a CC BY-NC license. Magnetic-field-temperature (B - T) phase diagrams at 2.3 GPa (b), 2.56 GPa (c), 2.68 GPa (d), 2.75 GPa (e), 2.9 GPa (f), and 3.2 GPa (g), respectively. Here, the acronym AF' stands for the field-induced anti-ferromagnetism. (h) The pressure-magnetic-field (P - B) phase diagram at the low temperature of 0.4 K. The data points denote the experimentally determined phase boundaries, and the regions enclosed by these boundaries correspond to distinct ground states. The red region is the T -linear specific heat phase, which is most pronounced around the pressure of the zero-field first-order PS-AF transition.

